

Microbiota

In damp conditions, the organic base of the plaster becomes a perfect breeding place and feeding site for a variety of bacteria, algae, coprophilous (dung-feeding) fungi whose spores attack the binder and penetrate deep into the substratum, secreting corrosive enzymes





Insects

The holes in the plaster left by biodeterioration are excellent sites for insect eggs, larvae and pupae. The majority of the damage to the Ajanta murals is done by silverfish (*Lepisma saccharina*), which feed on carbohydrates in the vegetal cellulose and live deep within the plaster. Since silverfish migrate up from the Waghora river, the use of pyrethrum (the toxin used in mosquito repellents) along the pathways has reduced the menace





Historic blunder

Major Robert Gill and John Griffiths, the two Victorian-era artists who produced exhaustive Ajanta copies, coated the murals with varnish. Ace Italian restorer Lorenzo Cecconi, in 1920, used unbleached shellac liberally. Both substances proved disastrous over time, cracking up the surface and changing the colours

MICROCLIMATIC INDICES

Relative humidity

A stable RH value of 55-60% at 25°C was found to be ideal for the paintings. High humidity is conducive to microlife. A NEERI study showed that tourist breath and body heat are big factors here. Ajanta's humidity also varies widely. In summer, the dry heat releases the moisture trapped in the clay plaster, leaving cracks in the pigment

Carbon dioxide

NEERI recorded peaks of 836 ppm, mainly on account of tourist breath. This has loosened the white pigment grains

Microbial load

The microclimate in the caves supports all kinds of microbes, another factor in the biodeterioration of the murals

Graphics by TANMOY CHAKRABORTY

rock. Across the rolling greens, a white Buddha, a speck from here, communes with the one who sits in dharma chakra mudra in Cave 1. The Tibetan pilgrims on that bus may not know it, but a homegrown monastic institution, of Ambedkarite persuasion, has taken up residence on those hills. Dhammachal, they call it.

Some things perish too. Up the first flight of stairs, a fork in the path leads to the other side of the river, the long walk. This affords a full, mid-range view of the spectacle, more of the grandeur in one frame, the falling cliffs with those chaitya porticos set into them, instead of entering edge-wise without perspective. A small compensation for missing the famous view from the top, as the first Englishman saw it. Here, amidst the chirping bulbuls, babblers and friendly locals selling PoP Buddhas, someone points to the bricks lying strewn amidst the bramble. Not debris…clearly laid in patterns, bigger than normal. Then the devastating news: it's an ancient burnt-brick monastery, perhaps once topped by carved wooden frames. Older than Ajanta? We don't know. A dig in 2002 had found a Byzantine coin here…of emperor Theodosius II. Then, incredibly, abandonment. Orphanhood. The land belongs to the forest department. The PWD has cut this path right *through* bricks that cry out for carbon-dating.

On the other side, just a minute away across the bridge from this dereliction, lies the cynosure of the world's eyes.

As one of the first four Indian sites to be inscribed on the Unesco World Heritage list in 1983—along with the Taj Mahal, Agra Fort and its own sister site Ellora—it receives far more international and thus local star treatment than the average Buddhist rock-cut cave site. The asymmetry is striking, but the attention itself is natural. Ajanta *demands* it of you: aesthetically, it's a complex jumble, your eyes don't know where to pause, what sensory input to prioritise, which feeling to soak in. The austerity of the chiselled rock stupa? Or the infuriatingly faded palimpsest of Jataka colours, the robes, the skin tones, the serene Buddha eyes, the mithuna couples, the chariot-mounted kings and yakshas, the teeming stories aching to reach you through the dotted lines of history? Who were these people who painted this banquet of life? You have read obscure names: the Mulasarvastivada sect, for one… the historian's nib is still paused mid-air.

By all accounts, it's in better shape today in some crucial respects than a few decades ago. For one, the lonely frontier sentinel of the Archaeological Survey of India (ASI)—a small set of valiant and possessed men—laboured away on rock and pigment, against supreme odds, with all the monastic devotion of the an-